Joshua (Stapley) Montoya

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PROFESSIONAL SUMMARY

Data-focused engineer with 10+ years of experience in analytics, systems engineering, and data workflow automation, with several years building and maintaining ETL pipelines that support cross-functional teams and business decision-making. Strong proficiency in Python and SQL.

TECHNICAL SKILLS

Data and Processing: Python, SQL, Dagster, Airflow, dbt, Pandas, DuckDB

Cloud & Infrastructure: GCP (GCS, BigQuery), AWS (S3, Athena, Glue), PostgreSQL, Docker, Terraform

PROFESSIONAL EXPERIENCE

Senior Systems Engineer — October 2015 – Present
L3Harris Technologies, Inc — *Salt Lake City, UT | Remote*

- Designed and automated end-to-end ETL pipelines using Python, SQL, and Dagster to ingest, validate, transform, and orchestrate data movement between analyst teams, replacing manual, UI-dependent workflows.
- Developed automated pipeline triggers and validation logic to detect new input datasets, verify steady-state conditions, and prevent redundant downstream processing.
- Modeled and transformed production datasets to support downstream analysis, improving data accessibility and reducing data preparation time by over 50%.
- Reconciled IDs between evolving schemas to automate data movement and eliminate redundant work across teams while logging noncritical mismatches for analyst review and resolution.
- Built batch ingestion workflows to migrate flat-file datasets into centralized databases, reducing manual processing time from minutes to seconds per record.
- Collaborated with cross-functional teams to define data requirements and support system-level data architecture and integration decisions.
- Led teams of analysts while mentoring junior engineers, coordinating technical efforts, and driving the successful delivery of customer-facing projects.

Intern, Advanced Engineering — May 2014 – August 2014
Teleflex, Inc — *Reading, PA*

- Developed a C-based prototype device for ingesting and processing real-time image data for catheter tracking.

SELECTED TECHNICAL EXPERIENCE

NYC Building Permits Mapping

- Built and orchestrated a batch ELT pipeline using Python, Airflow, dbt, Terraform, and Docker to ingest, load, and transform NYC building permit data into BigQuery for analytics.
- Developed an incremental dbt model to transform raw data in BigQuery and test the resulting output.
- Implemented partitioning, clustering, and chunked ingestion strategies to enable efficient querying and large-scale historical backfill processing.
- Built an AWS implementation using S3, Glue Data Catalog, Athena, dbt, Terraform, and Apache Iceberg, with incremental merges and automated data quality tests.

EDUCATION

Master of Computer Science — 2023
University of Illinois Urbana-Champaign

Bachelor of Science in Electrical Engineering — 2015
Brigham Young University